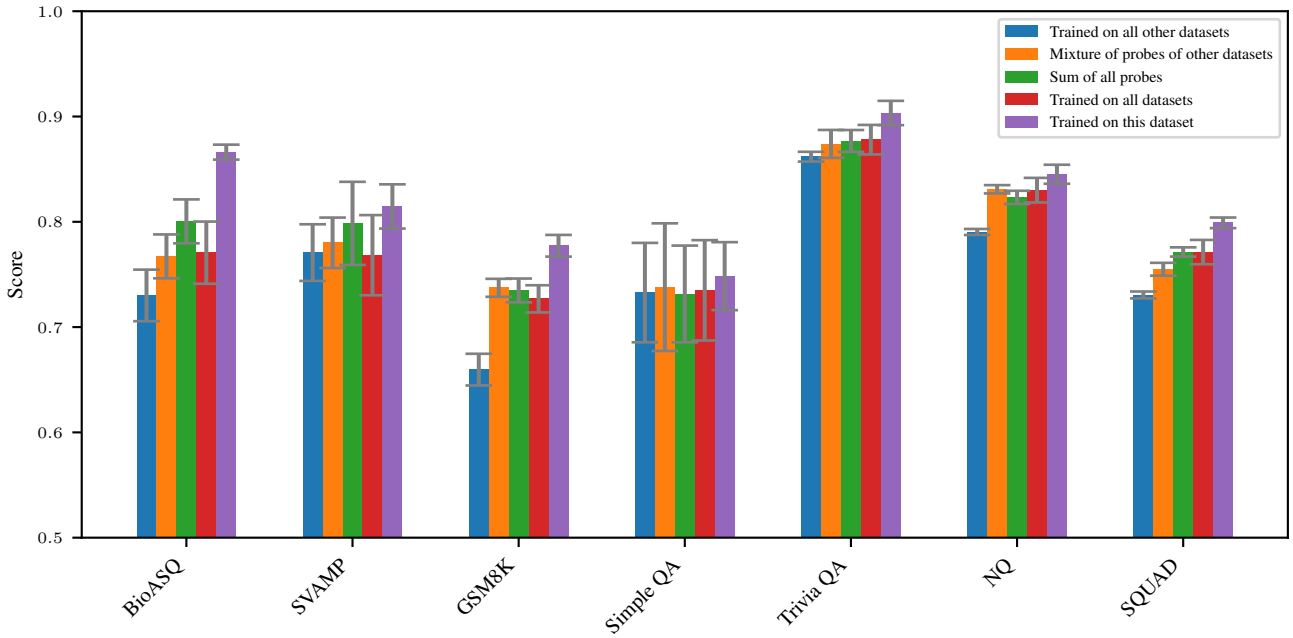


Figure 8. AUROC of linear probes at the stop token of the output in the multi-task setting, using L2 regularization.

In both cases, performance remains below that of linear probes trained directly on the target task. Moreover, the performance of this non-linear model matches that of simple linear probes trained on the same six held-out tasks.

These results show that even sophisticated probe architectures cannot bridge the performance gap with task-specific probes. The equivalence between complex and simple models trained on identical data suggests that the generalization failure stems from the orthogonal task geometries we identified, rather than limitations in model architecture.

4.3. Conservative approaches via conformal prediction

The previous sections have demonstrated that truthfulness representations are inherently task-dependent and fail to generalize across domains. This poses a critical challenge for real-world deployment: how can we maintain reliable uncertainty estimates when the distribution of user queries may differ from training data?

Given that guaranteeing generalization appears unattainable, we explore whether conservative calibration methods can provide reliability guarantees despite poor cross-task transfer. We focus on conformal prediction as a principled approach to control error rates, examining scenarios where avoiding false endorsement of incorrect information is crucial.

We consider the setting of Section 4 where probes are trained on multiple tasks and are evaluated on a new, unseen domain. Specifically, we train probes on all datasets except one test task using Equation (1) and seek to ensure that the false positive rate remains below a threshold $\alpha = 0.3$ on the held-out task. We compare three approaches: plain probes with default thresholds (Plain), standard split conformal prediction (Vovk et al., 2005) (CP), and a variant designed for multi-task settings (Park et al., 2022) (Meta-CP).

Given a trained probe $f(h) = \theta^\top h + b$ obtained by Equation (1), these methods calibrate a threshold τ such that the probe's confidence score must exceed τ before predicting an answer as correct. For new hidden states $h_{t,\ell}$ corresponding to question-answer (q, a) with label y, the following bound on the false positive rate holds:

$$\mathbb{P}\left(f(h_{t,\ell}) > \tau \,|\, y = -1\right) \leq \alpha\,.$$

This ensures that, on average, the false positive rate will be at most α for new questions. We refer to Vovk et al. (2005) and Park et al. (2022) for methodological details. For the multi-task variant, we set both hyperparameters to 0.3 and artificially randomly split calibration tasks into subtasks of size 1000 to match the experimental setup from Park et al. (2022).

The results in Figures 10 and 11 reveal substantial differences between approaches. Plain probes achieve a mean false positive rate of 0.34, exceeding the target threshold of 0.3, with high variability (80th percentile: 0.69). Standard conformal prediction reduces the mean false positive rate to 0.25, approaching but not consistently achieving the target, with the 80th percentile still reaching 0.47. The multi-task variant achieves the strongest false positive rate control,

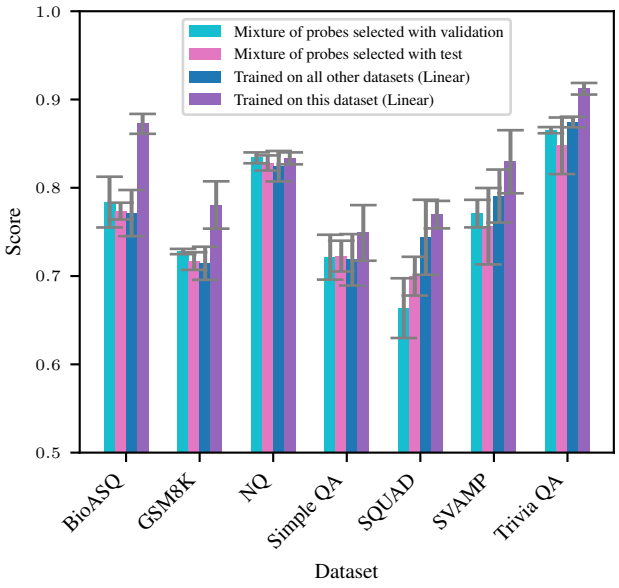
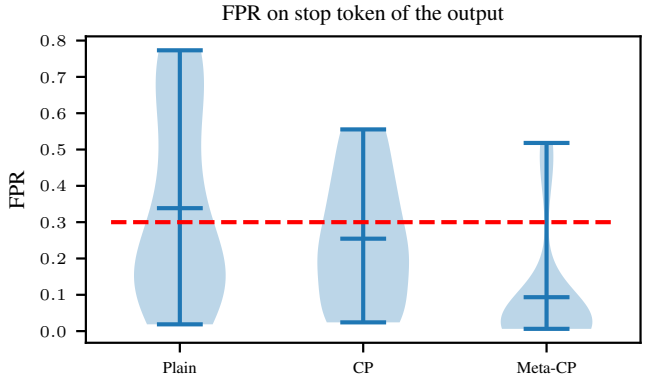
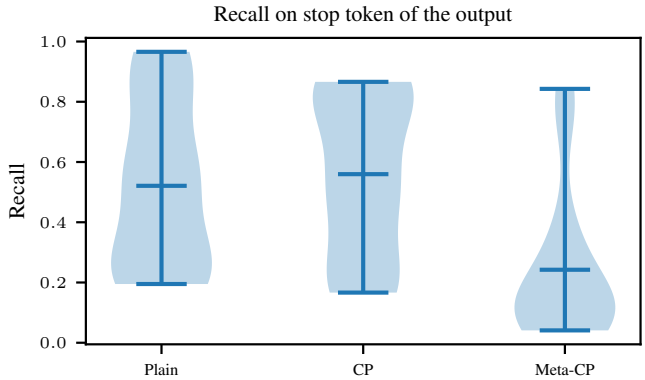


Figure 9. AUROC of different methods in the multi-task setting. Mixture of probes are trained on six non-target tasks and evaluated on the target task. "Validation" uses hyperparameters selected on a validation set from the training tasks; "Test" uses hyperparameters selected on the target task (oracle setting). Linear probe baselines ("Trained on all other datasets" and "Trained on this dataset") reproduce results from Figure 8 for comparison. Results averaged over 3 runs.



(a) Violin plot of the false positive rate (FPR) of the different methods with threshold $\alpha = 0.3$.



(b) Violin plot of the recall of the different methods with threshold $\alpha = 0.3$.

Figure 10. False positive rates (FPR) and recall for thresholds tuned using different methods: standard training (Plain), split conformal prediction (CP), conformal prediction for multi-task settings (Meta-CP). The results are averaged over 5 repetitions and test tasks.

with a mean of 0.09 and 80th percentile of 0.23, successfully staying below the target threshold. However, these improvements in false positive rate control come at substantial cost to recall. While plain probes achieve 0.52 mean recall and standard conformal prediction reaches 0.56, the multi-task variant falls to just 0.24. This dramatic reduction means that this conservative approach correctly identifies only about one-fourth of true positives, illustrating the fundamental trade-off when truthfulness representations fail to generalize. Moreover, this poor performance also reflects a deeper issue: conformal prediction assumes the scoring function $f(h)$ reliably ranks correctness across domains. However, our findings show that truthful and untruthful generations are only linearly separable within task-specific subspaces that vary significantly across tasks. When probes encounter out-of-distribution tasks, they become misaligned with actual correctness labels. To maintain the required false positive rate guarantees, conformal prediction must set extremely conservative thresholds, filtering out many correct answers not because they are ambiguous, but because the probe's direction no longer matches the task's geometry of truth. Our insight is that conformal prediction becomes overly conservative precisely because the underlying geometry fails to generalize.

Method	Mean FPR	Q-80% FPR	Mean Recall
Plain	0.34	0.69	0.52
CP	0.25	0.47	0.56
Meta CP	0.09	0.23	0.24

Figure 11. False positive rates (FPR) and recall for thresholds tuned using different methods: standard training (Plain), split conformal prediction (CP), conformal prediction for multi-task settings (Meta-CP). The results are averaged over 5 repetitions and means and 80% quantiles (Q-80%) are considered over test tasks.

Conclusion. The premise of the "geometry of truth" hypothesis is that one might be able to detect whether an LLM provides a correct answer when prompted with a question. These works claim that a simple classifier, taken as inputs the activations generated by the LLM as it produces its answer, can suffice to predict the correctness of the final

answer. Although extremely appealing, we show in this work that such a promise may not be yet reliable enough as it fails to transfer across domains and tasks, notably if the domain on which the probe was trained is markedly different from that where the performance of the classifier is evaluated. We explain this failure to generalize by noticing that probes trained independently on various tasks have both low similarity and small feature overlap when trained with sparse regularizers. We have explored more advanced classification paradigms, such as mixture-of-probes, which could have been able to handle this heterogeneity, but we were not able to achieve reliable generalization. We conclude that LLMs likely have multiple geometries of truth, but that they are irreconcilable and highly task-dependent.

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